

Creativity - ERROR 404: Navigating creativity and ownership in the age of ai-assisted students.

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Generative AI (Gen-AI) tools have completely overhauled education on every level, enabling students across different ages to boost their creativity and enhance learning. However, multiple questions are being raised with regards to the usage of such tools in creative tasks. This study examines the role of human-AI collaboration in fostering creativity among undergraduate students. Through this study we will understand and disseminate the role of human-AI collaboration with respect to fostering creativity amongst undergraduate students by synthesizing information and insights obtained from qualitative interviews of 20 undergraduate students studying in design and engineering disciplines in metropolitan Indian universities. Some of the key themes have emerged during this process: AI as a creative collaborator, ethical concerns, balancing originality with AI assistance, challenges in identifying AI-generated content, and the impact of Gen-AI tools on creative independence. we have made an effort to inform the development and conceptualization of such tools through these qualitative insights while addressing various ethical and motivational challenges..

CCS CONCEPTS • Human-centered computing → Human computer interaction (HCI) → User studies - Human-centered computing → Empirical studies in HCI - Human-centered computing → HCI design and evaluation methods - Human-centered computing → Collaborative and social computing → Collaborative and social computing theory, concepts and paradigms - Social and professional topics → Computing education → Student assessment - Social and professional topics → Codes of ethics

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1 INTRODUCTION

The fast inclusion of generative AI tools into learning environments has transformed how students engage with creative and academic work. Tools such as ChatGPT for text generation and DALL-E for image creation, provide unparalleled opportunities for increasing creativity, refining workflows, and improving education. Generative AI, particularly in creative domains, has the ability to empower students by creating new channels for self-expression and idea inception. However, such breakthroughs raise serious concerns about creative freedom, originality, and the intermixing of human and machine creativity (Habib et al., 2023). There is the danger this poses to student independence and ownership (Kim & Maher, 2023; Berni et al., 2024; Yun et al., 2022). Students perceive both risk and opportunity: increased efficiency and diversity of ideas but also dependence, blurring of boundaries between work by humans and work by the AI, as well as concerns around the ethics within coursework (Zhou et al., 2024; Dakakni & Safa, 2023; Chan & Hu, 2023). These tensions are sharp where originality and voice are key goals, and where AI's role changes by task—from ideation and visualization to refinement, translation, and technical support (Kim & Maher, 2023; Berni et al., 2024; Yun et al., 2022).

HCI design writings point out that the outcomes depend on when and how you integrate the AI—as the stage of the task, student proficiency, and collaborative definition—more than on the presence of the tool itself (McCormack et al., 2020; Holstein et al., 2019). Scholars working on creative education **stress** scaffolds that preserve authoring and agency, such as process reflection, transparency around the use of the AI, and clear limitations on where the AI is acceptable in the work flow (Overono & Ditta, 2023; Lim et al., 2023; Chan & Hu, 2023). Qualitative chronicles of how students themselves balance identity, independence, and authorship amid everyday human–AI collaboration are still scant, despite widespread debate, particularly across India's engineering and design courses (Habib et al., 2023; Zhou et al., 2024).

The current study explores the noted lack through semi-structured interviews with undergraduate designers and engineers, examining variation by task type and proficiency of AI engagement, the strategies students use to set boundaries toward preserving their creative identity, and the expression of ethical considerations within customary coursework (Habib et al., 2023; Chan & Hu, 2023). Following the concerns raised by respondents and by design education pedagogic advice, this paper suggests a Compact Framework of Ethical Integration Principles—intent and stage-fit, boundaries of authorship, disclosure and attribution, ownership duty, and reflective documentation—to support educators and students in maintaining their agency while working effectively with AI in design studio settings and assignments (Overono & Ditta, 2023; Holstein et al., 2019; Lim et al., 2023).

Contributions

1.1 Contributions

- First, we present qualitative themes concerning how undergraduates relate to creativity, independence, and ownership with generative AI in design and engineering programmes (Habib et al., 2023; Chan & Hu, 2023).
- Second, we employ a task-based approach to analyze the functions AI can take (assistant, partner, critic) during ideation, refinement, translation, technical support, and aesthetic exploration, mapped to student proficiency gaps (Kim & Maher, 2023; Berni et al., 2024; Yun et al., 2022). Ethics-integration practice-oriented principles, including setting boundaries, disclosure, ownership, and reflective process, provide evidence for arts education (Overono & Ditta, 2023; Holstein et al., 2019; Lim et al., 2023).
- Finally, we outline pragmatic considerations for design education and appraisal that support creative identity when using artificial intelligence in initial idea generation and subsequent refinement procedures (Overono & Ditta, 2023; Kim & Maher, 2023)...

2 RELATED WORK

2.1 Generative AI in Education

During the current scenario, generative AI has been revolutionizing education by providing digital services or features, some of which are assessment, personalized learning support, as well as intelligent tutoring systems, with Chat-GPT by OpenAI being the leading tool based on its extensive use (Bahroun, Anane, Ahmed, & Zacca, 2023). Such technologies have been proven to show a significant impact within specific knowledge domains by showing an improvement in learning, especially in creative fields (Mishra, Warr, & Islam, 2023). For example, DALL-E has abilities that can be beneficial for students in visual arts by generating unique images that are based on their descriptions, and ChatGPT can provide assistance in the form of instant feedback on student work or creative writing prompts (Bahroun et al., 2023; Kieser, Wulff, Kuhn, & Küchemann, 2023; Adiguzel, Kaya, & Cansu, 2023). Apart from the creative realm, generative AI language models also have the potential to transform geography and environmental education through quick feedback offered to students in a self-paced environment (Chang & Kidman, 2023) as well as quantitative reasoning tasks, which are prominent in physics and concept tests, thus leading to enhancements in educational research (Kieser et al., 2023) and learning and educational processes, which benefit stakeholders, primarily teachers and students, while promoting responsible and ethical use (Adiguzel et al., 2023). Generative AI (GenAI) in education offers opportunities for personalized learning and assessment but raises concerns about academic integrity and student dependency (Forment et al., 2024; Kars, 2023). While GenAI can enhance teaching methods and reduce workload, it challenges traditional educational paradigms and necessitates ethical considerations (Forment et al., 2024; Walczak & Cellary, 2023). The technology's impact on creativity and originality in student work is a significant concern, requiring strategies to maintain academic standards (Forment et al., 2024; Kars, 2023).

2.2 Generative AI's Impact on Creative Independence in Education

Creative independence and originality in educational psychology are complex concepts that have evolved over time. Creativity, once attributed to divine sources, is now recognized as a multifaceted human capacity involving cognitive, conative, and environmental factors (Morska & Storchova, 2024). It is considered an important component of problem-solving, cognitive abilities, and overall well-being (Plucker et al., 2004). The educational psychology of creativity focuses on understanding, measuring, and developing creative thinking and problem-solving skills in all students, not just the gifted (Houtz, 2003). Despite its importance, creativity research in educational psychology faces challenges due to myths and stereotypes, stemming from a lack of precise definitions (Plucker et al., 2004).

Studies explore the impact of AI tools on students' perceptions of creativity and learning. Students generally view AI positively for autonomous learning and academic writing, recognizing its benefits in grammar checks, plagiarism detection, and essay outlines (Laínez Quinde et al., 2024; Malik et al., 2023). However, concerns exist about AI's potential effects on creativity, critical thinking, and ethical writing practices (Malik et al., 2023). Research indicates that students with higher AI understanding report more positive attitudes towards its classroom integration, while those with limited knowledge express fear (Marrone et al., 2022). Although AI can support creative thinking, it may also negatively impact creativity and creative confidence (Habib et al., 2023). Most students believe AI cannot match human creativity (Marrone et al., 2022). To address this, researchers emphasize the need for clear operational definitions and a nuanced understanding of creativity's varied dimensions in educational contexts (Morska & Storchova, 2024). Future research should focus on expanding the literature base, employing mixed methods, and developing new tools for assessing creativity in educational settings (Huang & Alhassan, 2024).

2.3 Student Perspectives on AI-Generated Content and Authorship in Creative and Academic Contexts

University students engaged with generative AI for creative writing, demonstrating emerging AI literacies and reconceptualizing creativity as distributed across human and non-human agents (Tsao & Nogues, 2024). Middle schoolers' interactions with text-to-image AI models revealed novel digital artifacts and collaborative discussions, with teachers facilitating critical reflection on authorship and copyright (Vartiainen et al., 2023). Marketing communications students predominantly viewed AI as a tool rather than a co-creator, expressing concerns about overreliance potentially eroding creative skills (Karimova et al., 2024). In team settings, design students collaboratively used GenAI for creative tasks, perceiving it as an efficient tool rather than a true collaborator (Han et al., 2024). Students often struggle to distinguish between their own work and AI-generated output, leading to unintentional plagiarism and challenges in claiming authorship (Vaccino-Salvadore & Hall Buck, 2021). This issue is exacerbated by the use of AI writing assistants, which create a mental dilemma as users tend to credit AI more for creative tasks (Azmine Tushik Wasi et al., 2024).

The "AI Ghostwriter Effect" reveals that users don't perceive ownership of AI-generated text but still declare themselves as authors, with personalization having little impact on this effect (Draxler et al., 2023). In educational contexts, students engage with AI tools for creative writing, decentering authorial attribution and reconceptualizing creativity as distributed across human and non-human agents (Tsao & Nogues, 2024). The use of generative AI in academic writing raises complex issues of authorship, ownership, and academic integrity, necessitating transparent reporting and new mechanisms to identify AI use (Bozkurt, 2024).

2.4 Working together with AI on creative tasks

AI has become increasingly integrated into a variety of creative fields including design and content creation. AI systems can cooperate with humans in real-time to produce creative results, performances, and artifacts. These collaborative AI systems facilitate spontaneity, performance, and co-creation through continuous, moment-to-moment interaction. Supporting trust, cooperation, shared awareness, and a shared information space are all important factors when creating such systems (McCormack et al., 2020). AI-based co-creative systems in design, for example, can improve creativity by facilitating the discovery of innovative design solutions during the early idea generation phase. These systems can provide inspirations based on conceptual similarity, significantly influencing the uniqueness, diversity, and quantity of ideas produced (Kim and Maher, 2023).

One of the biggest challenges in amalgamating AI into creative tasks is maintaining a consistent equilibrium between utilizing AI's capabilities and maintaining students' creative independence. While AI can augment human creativity by providing novel inspirations and enabling the exploration of new ideas, it is crucial to ensure that students do not become overly dependent on AI tools. For instance, in the context of empathic conversations, peer supporters were able to use AI feedback without becoming completely reliant on it, suggesting that AI might enhance human creativity without undermining it (Sharma et al., 2022). The design of AI systems should prioritize human-centric approaches, where AI acts as a tool to augment rather than replace human creativity (Anantrasirichai and Bull, 2020). Additionally, understanding the scope and perception of two-way communication between human and machine agents is vital to support reflection on conflict, error, evaluation, and flow in creative processes (McCormack et al., 2020).

2.5 Design ideation with the AI

Early-stage design-AI collaboration studies suggest generative systems are able to broaden idea spaces, augment fluency, and provide novel combinations but risk converging toward customary or tool-shaped patterns without scaffolding (Berni et al., 2024). Text-to-idea pipeline research suggests large-language models are inspiration seeds triggering divergent

exploration with measurable end-states on uniqueness, diversity, and concept count for the design's front-end (Yun et al., 2022; Kim & Maher, 2023). Complementary HCI work on co-creative tools and timely points of human–AI interaction signal that outcomes are stage-fit, transparency, and role choreography toward assistant, partner, or critic—rather than by tool presence itself (McCormack et al., 2020; Holstein et al., 2019). As one body, this work positions AI as an effective but delicate ideation partner whose end-states are contingent on task definition, user expertise, and process guidance (Kim & Maher, 2023; Berni et al., 2024).

But the majority of past work focuses on professional designers or orchestrated ideation labs, emphasizing professional practice, reviews of tools, and lab-based assessments of idea novelty and variety (Berni et al., 2024; Kim & Maher, 2023). The present study, however, looks at undergraduate students from design and engineering backgrounds, examining lived, course-level usage of AI during coursework and projects, where questions of identity, authoring, and boundaries around assessments are salient. A student-centric, cross disciplinary point of view by necessity reveals how gaps around proficiency and academic culture shape when the roles of the AI are assistant versus partner, how boundaries are determined to preserve authoring, and how ethical concerns manifest in everyday work—including detail less obvious from studies focusing on designer-only ideation labs (Yun et al., 2022; Holstein et al., 2019; McCormack et al., 2020). By grounding analysis on student practice rather than professional studios, the study generalizes the results from ideation studies to learning contexts and grounds education-focused policy and practice on the responsible design-to-use of AI during early-stage creativity work (Kim & Maher, 2023; Berni et al., 2024).

3 METHODS

3.1 Overview and Flow

The study employed a multi-stage qualitative method consisting of: recruitment → screening based on frequency and proficiency → semi-structured interviews → transcription → collaborative coding in Figma Jam → theme development → validation checks. This process ensures straightforward sampling, multiple forms of data collection, and rigorous, auditable qualitative analysis that conforms to current practice in HCI and design studies (Braun & Clarke, 2006; Guest, Namey, & Chen, 2020).

3.2 Participants and Sampling

Twenty undergraduate design and engineering students from India were recruited through purposive sampling to ensure variation by discipline and intensity of AI use. Fourteen interviews took place online and six in person to accommodate availability and ensure contextual depth from the in-person sessions. "Frequent generative AI usage" was defined as the use of one or more generative tools on at least three days per week for at least three continuous months. Screening recorded usage bands: weekly (3–4 days/week), daily (5–6 days/week), and multiple times daily (two or more sessions/day on five or more days/week). Screening also noted the primary tools used (e.g., ChatGPT/GPT-4 class models for text; DALL·E, Stable Diffusion, and Midjourney for visualizations) and self-rated proficiency from 1 to 5 (1 = novice, 5 = expert), which were aggregated to describe the variation by expertise. This methodology operationalizes frequency and proficiency in a manner similar to previous HCI sampling methods for AI-mediated creative work while maintaining ecological validity (Guest et al., 2020).

Table 1. Participant demographics and AI use

ID	Discipline	Primary AI tools	Self-reported proficiency (1–5)	Usage band
P1	Computer Science	ChatGPT, GitHub Copilot	5	Daily
P2	Mechanical Engineering	ChatGPT, Midjourney	3	Weekly
P3	UX Design	DALL·E, ChatGPT	5	Multiple-times-daily
P4	Electrical Engineering	ChatGPT, Bard	3	
P5	Computer Science	GitHub Copilot, ChatGPT	4	Daily
P6	Industrial Design	Midjourney, DALL·E	3	Weekly
P7	Computer Science	ChatGPT, Claude	5	Multiple-times-daily
P8	Visual Design	DALL·E, Stable Diffusion	4	Daily
P9	Mechanical Engineering	ChatGPT	2	Weekly
P10	Computer Science	GitHub Copilot, ChatGPT	3	Daily
P11	Electrical Engineering	ChatGPT	2	Weekly
P12	Visual Design	Midjourney, Stable Diffusion	4	Daily
P13	Electrical Engineering	ChatGPT, Bard	3	Weekly
P14	Industrial Design	DALL·E, Midjourney	4	Multiple-times-daily
P15	Computer Science	ChatGPT, GitHub Copilot	5	Daily
P16	Interaction Design	ChatGPT, Midjourney	3	Weekly
P17	Mechanical Engineering	ChatGPT, Claude	3	Daily
P18	Product Design	Midjourney, Stable Diffusion	4	Daily
P19	Computer Science	Chat-GPT	2	Weekly
P20	Electrical Engineering	Bard, Chat-GPT	3	Daily

3.3 Interview Protocol

Semi-structured interviews investigated the extent to which students employed AI throughout the creative process, their perceptions of effects on creativity and independence, and sketched boundaries around authorship and originality. The key protocol follows below, with the full guide and probes published in Appendix A. The interviews commenced with context (discipline, semester, usual coursework), followed by an exploration of recent uses of AI episodes with task-specific probes, e.g., "Walk through the most recent time that AI was utilized for ideation—what prompt(s), iterations, how the idea changed, and what was edited out or discarded?" "Explain an artifact/show an artifact and label what is AI versus humankind's," and "Explain any disclosure to the instructors, and rationale." The probes were crafted to encourage process evidence and reflection instead of opinions about the tools, by following qualitative interviewing best practices for work that is creative (Braun & Clarke, 2006). The sessions took 35–45 minutes, depending on the participant's pace, were audio-recorded with permission, and transcribed word for word.

3.4 Transcription and Data Management

All interviews were transcribed and gently cleaned of disfluencies, retaining meaning-critical phrasing. Offline interviews also followed the same procedure but were carried out in silent rooms on campus. Participant identifiers were pseudonymized; any institutional or personal identifiers were removed from the transcripts before they were analyzed. Metadata tables captured interview mode (online/offline), duration, discipline, proficiency rating, and prevailing tools utilized to allow for descriptive analysis as well as cross-case comparisons.

3.5 Task taxonomy

To put the analysis into concrete practices, the research provided an activity-based taxonomy based on the respondents' stories:

- Ideation: conceptual divergence, analogy-based cues, and prompt-based exploration.
- Refining/elaboration: clarification, improvement in structure, or addition to an idea or first version.
- Translation/reformatting: translating content from one modality/format to another (e.g., from notes to outline, from outline to storyboards).
- Technical assistance: code pieces, data wrangling, layout suggestions, or toolchain problems.
- Aesthetic exploration: variations on style, mood boards, visual iterations.
- Concept synthesis: combining AI suggestions and personal reasoning into an integrated direction.

3.6 Analysis approach

Collaborative qualitative analysis was conducted in Figma Jam, allowing for shared visual coding, clustering, and iterative development of themes. The process followed reflexive thematic analysis: familiarization, open coding, axis development and clustering, identification of candidate themes, with review and definition thereafter (Braun & Clarke, 2006). Coders created sticky-note excerpts with source IDs and line references, added initial inductive codes, and increasingly structured clusters within task categories, perceptions of creativity impact, boundaries to authorship, and disclosure practices. Within the canvases of Figma Jam, a living codebook was kept, including code names, terse definitions, inclusion and exclusion criteria, and exemplar quotes. Merging and renames of codes were also tracked in a versioned change list to trace the development of the codebook. To maintain analytic rigor equivalent to dedicated CAQDAS environments, the team took

frame-level snapshots to indicate milestone states, followed a consistent annotation convention, and made use of color-coding to distinguish descriptive codes, process codes, and evaluative judgments.

3.7 Reliability and validation

- Analyst triangulation: Two researchers made independent coding of a preliminary small group of transcripts (25%) and cross-checked differences to reconcile code definitions, subsequently made intermittent calibration checks on new interviews (Guest et al., 2020). Where single coding was conducted on later transcripts, second-pass spot checks together with memo-based auditing were utilized to maintain consistency.
- Negative case study: Explicitly searched for were contradictory instances (e.g., students reporting increased originality following AI use versus reporting convergence or dependency), and incorporated within the boundaries of themes.
- A saturation memo was run to note the appearance of new codes; during the last three interviews, no new substantive code was incorporated, confirming thematic saturation by $n = 20$ for this breadth (Guest et al., 2020).

3.8 Ethics

All participants provided consent before data collection. Participation was voluntary, and right to withdraw was allowed at any moment. No compensation-biased participation other than tokens of appreciation typical to student research. Data had been stored on secure, access-restricted drives; only aggregated, anonymised results are presented.

4 RESULTS

Twenty undergraduate participants (14 online, 6 offline) identified how generative AI affected creativity, independence, and authorship through repeated themes mapped to a task taxonomy: ideation, elaboration/refinement, translation/reformatting, technical assistance, aesthetic exploration, and concept synthesis. More skilled students most often described the role of AI as partner or critic during early-stage ideation and end-stage synthesis, while less skilled students most often utilized AI as an assistant during late-stage refinement, translation, and routine problem-solving.

4.1 Perceptions of creative independence

Students characterized a double-edged dynamic: AI would broaden ideas and speed up advances, but over-dependence threatened to stifle originality and personality. One respondent commented, *"If I apply AI under narrow boundaries, it actually enhances creativity. The suggestions made by AI complement my creative thoughts. However, when I depend on it too much to conform to deadlines, creativity declines."* Another reflected, *"I apply AI to give ideas on coding projects. Although it often generates general ideas, saving me time, which allows me to polish the outputs."* A dissenting voice cautioned, *"When depending on AI, my thinking ends where the suggestions by the AI cease. It overwhelms the ability to develop deepened ideas."* Higher-proficiency respondents had the tendency to limit the role of AI to early divergence and late critique, while lower-proficiency respondents resorted to the efficiency of task completion by leveraging AI.

4.2 Self-expression, boundaries, and identity

Participants underlined the need for setting boundaries to protect authorship and authenticity, particularly where visual and narrative work is created. One participant encapsulated, *"So long as you remain faithful to the original, say, with photography, and you're not changing so much to make the picture appear unrealistic, you still retain your signature."* Psychological tensions arose around ownership and worth: *"If I'm too dependent on the AI, the outcome does not feel mine. The fault lines and humanness are what creativity is all about, and the AI will never reproduce that."* One alternative perspective flagged the idea of empowerment alongside holding back: *"I utilize the AI for generating ideas when creativity stalls, but always make sure that what I submit is my own work product and interpretation"*. Higher-proficiency students articulated specific guardrails (e.g., what AI is allowed to touch), while novices spoke more generally about caution.

4.3 Collaboration dynamics and role variability

AI's role shifted by task and discipline—from assistant in technical work to collaborator for creative visualization. One student noted, *"Working alongside the AI on code, I conceptualize the big picture and planning of what I'm working on, but the AI assists with the repetitive process, such as typing out routine syntax, which is slow to remember."* Another noted overdependency: *"It's too tempting to overuse the AI too much, but sometimes you have to push yourself to make the brain work. While the AI is valuable, overdependency will make us lazy as well as slow our ability to creatively solve problems, so one must balance assistance with independent thinking."* Creative environments emphasized co-creation: *"The biggest challenge for me is conceptualizing scenes from the storyboard. Instead of sketching out by hand, I experiment with different tools within the AI to recap what is visualized, so that I'm able to hone and firm up the storyboards for the comic book."* A negative example emphasized bypassed learning: *"We just put the assignment up on ChatGPT, and we'll just get all the answers. The learning here is zero."* Some noted synthesis: *"AI's input, but I work it alongside what's in my brain to build something new and inventive."*

4.4 Boundaries for independence

Students are often assigned personally meaningful activities to manual work while utilizing artificial intelligence for routine augmentations. As one respondent put it, *"If I'm enthusiastic about something, I believe that I'll be able to do it by myself. If it's something enjoyable, then I'll ensure that I get the maximum work out of it."* Comments related to efficiency trade-offs were provided: *"Work has become so much simpler, and I don't work the same way anymore. Although hard work still happens, the intensity of the work is lowered marginally."* Some respondents set rigid boundaries where authenticity's key domains lie: *"I won't create things that I'm capable of capturing, such as wildlife photography. Those things where the physical or creativity-based presence is necessary are strictly mine alone."* Oversight still took precedence: *"AI is the servant, not the boss. I make sure I know what the output is before putting any trust in it, and if something appears ambiguous, I give the additional input to make the end result better."* Higher-proficiency respondents set explicit boundary rules, while lower-proficiency respondents expressed intentions but not systematic strategies

Table2. Results Summary

Theme	Concise definition	Representative quote	Prevalence (n out of 20)	Illustrative task types
Perceptions of creative independence	AI boosts fluency and speed but can suppress originality when over-used	“If I use AI within a limited scope, it can actually boost my creativity... when I rely on it excessively... my originality decreases.”	17	Ideation; Elaboration/refinement
Self-expression, boundaries, identity	Students set limits so AI supports, not replaces, their voice	“As long as you stay true to the original... you still maintain your identity.”	15	Aesthetic exploration; Concept synthesis
Collaboration dynamics and roles	AI shifts between assistant, partner, critic by task and proficiency	“I handle the overall structure... AI assists with repetitive tasks...”, “We just upload the assignment... the learning here is zero.”	18	Technical assistance; Ideation; Translation
Boundaries for independence	Deliberate guardrails preserve authenticity and ownership	“If I’m passionate... I’ll do it on my own.”; “AI is a servant, not the boss.”	16	Concept synthesis; Aesthetic exploration; Elaboration

5 DISCUSSION

Findings indicate that generative AI can productively expand ideation, accelerate refinement, and lower technical friction, but unguarded use risks convergence, dependency, and dilution of authorship. For design education and studio practice, the most effective pattern observed is staged, scaffolded integration: permit AI for early divergent exploration and mechanical transformation; require human-led concept synthesis, rationale, and final authorship; and institute lightweight provenance to keep contributions transparent. Practically, this translates into assignment designs that separate ideation from generation, require reflection logs to document boundaries and decisions, and adopt clear disclosure policies matched to task and objectives. In studios, instructors can employ scaffolded prompts that (a) focus early AI use on breadth, analogy, and reframing; (b) mandate human synthesis notes explaining what was accepted, adapted, or rejected and why; and (c)

assess not only outcomes but also process evidence demonstrating authorship and critical judgement. For critiques, a short, standard provenance statement can specify tools, stages of use, and human contributions, while rubrics add a line item for ownership claims aligned to contribution type (conceptual, structural, technical, stylistic). Program-wide, courses can publish a transparency policy with default disclosure modes per assignment type, and adopt shared reflection templates that ask for benefits, risks, boundaries, and attribution plans; this normalizes ethical use without chilling legitimate, learning-aligned experimentation. Finally, proficiency-sensitive scaffolds matter: advanced students benefit from partner/critic roles in ideation and late-stage critique, whereas developing students gain most from assistant roles in refinement and translation coupled with explicit guardrails and verification habits. Across these choices, the aim is consistent: preserve creative identity and learning while channeling AI's affordances toward exploration, craftsmanship, and clear authorship.

5.1 Limitations

This study uses one national setting and one deliberate sample of design and engineering undergraduates, so generalizability to other territories, disciplines, or institutional settings is potential but limited; the qualitative breadth is conducive to depth rather than breadth, and the self-reports will tend to under- or over-estimate actual use or report social-desirably practices; finally, tool collections are extremely fast-changing environments, so some usage patterns and perils will depend on model abilities and institutional policy.

5.2 Future Work

Future work Three extensions are given first priority: one, extend and diversify samples over regions, institutional types, and artistic and technical disciplines to study cultural and curricular moderators; two, take up mixed-methods by complementing interviews with simple quantitative measures like creativity self-efficacy scales, short originality/fluency exercises, and log-coded usage rates over task phases to cross-validate patterns; three, conduct longitudinal designs to study durability of effects on independence, originality, practices of authorship, and skill development over semesters as tools and policy mutually evolve.

6 FRAMEWORK FOR ETHICAL AI INTEGRATION

Concise, user-friendly guide that translates the findings of the study to useful recommendations for classroom instruction. It can be a standalone section following the Discussion or the final portion of the Discussion if there's little space. It comes with a compact table and ownership chart to enable fast insertion in course schemes, briefs, and marking schemes.

Core principles:

1. Context first: Frame AI application to learning goals and disciplinary protocols.
2. Transparency as default: Select a suitable disclosure mode per task.
3. Sustain authorship: Establish human versus AI roles based on contribution type and task stage.
4. Think long term: Think about skill development, identity, and field-level effects.
5. Document selections: Note the ethical rationale supporting AI application.

Table3. Framework Components

Component	What it is	How to apply	Stakeholders
Contextual appropriateness assessment	Quick screen aligning AI use with goals, norms, attribution, and impact	Use a 4-question check at brief and critique: goal, norms, attribution, stakeholder effects	Educators, students, program leads
Learning objective consideration	Tie AI scope to whether the goal is skill formation, exploration, or production	Tighten AI for skill formation; allow early-stage AI for exploration with reflection	Curriculum designers, instructors
Attribution expectations	Clear rules for acknowledging AI involvement	Require a provenance note and example templates in briefs	Students, instructors, integrity officers
Disciplinary norms	Field-specific conventions about technological support	Calibrate by domain (e.g., stricter for authorship-heavy studio; flexible in prototyping)	Department chairs, studio leads
Stakeholder impact analysis	Consider effects on peers, grading, and future work	Add a one-paragraph reflection at submission	Students, instructors
Transparency continuum	Match disclosure mode to task and goals	Default mode per assignment; allow justified deviations	Students, policy makers
Complete documentation	Step-by-step process record	Use in process-focused, formative assignments	Students, assessors
Summary attribution	Brief acknowledgment line	Use in outcome-focused creative work	Students, audiences
Implied support	Tacit recognition of common tools	Use where AI aid is normalized and low-stakes	Professional communities
Personal use disclosure	Private reflection on AI use	For exploratory study and practice logs	Individual learners

Component	What it is	How to apply	Stakeholders
Ownership responsibility matrix	Map human/AI claims by contribution type	Students label contribution claims; evaluators check consistency	Students, instructors
Long-term impact consideration	Anticipate skill and identity effects	Include short prompts in reflection	Students, educators
Ethical decision documentation	Mini-template to justify AI use	Require five fields: context, benefit, risk, boundary, attribution	Students, instructors

Table 4. Ownership Responsibility Matrix

Contribution type	Human-led	Human–AI collaboration	AI-led with human direction	AI-generated
Conceptual (intent, narrative, core idea)	Full human ownership	Primary human ownership	Shared ownership	Limited human claim
Structural (outline, composition, logic)	Full human ownership	Shared ownership	Limited human claim	Minimal human claim
Technical (code, formatting, tool steps)	Full human ownership	Shared ownership	Limited human claim	Minimal human claim
Stylistic (voice, visual style, tone)	Full human ownership	Primary human ownership	Shared ownership	Limited human claim

7 CONCLUSION

This study examines the interplay between generative AI tools and undergraduate students' creativity, highlighting the balance between collaboration and maintaining creative independence. The findings reveal that while AI enhances productivity, sparks ideas, and aids in routine tasks, it can also challenge creative authenticity and psychological well-being. Students view AI as both an enabler of innovation and a potential inhibitor of originality, emphasizing the importance of setting boundaries to preserve their personal identity in creative outputs. Collaboration with AI varies across domains, serving as an assistant in technical tasks and a creative partner in artistic endeavors. However, concerns about over-reliance and diminished effort underline the need for thoughtful engagement. The study underscores the dual role of AI in education, advocating for its use as a complement to human creativity rather than a replacement. These findings call for educators and policymakers to establish guidelines that support creative autonomy while leveraging AI's potential. Future research should explore broader contexts and long-term effects of student-AI collaboration, ensuring that AI's integration in education empowers students without compromising human innovation.

AI Disclosure

Generative AI tools were used to assist with language editing, copyediting, and drafting refinements (e.g., clarity, grammar, formatting consistency, and minor rephrasing). All study design decisions, recruitment and screening, interview protocols, data collection, coding/analysis, theme development, framework formulation, tables/figures, and final interpretations were conducted and verified by the authors.

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A APPENDICES

A.1 Interview guide and probes

Opening, consent, and context (3–5 min)

- Confirm consent to audio-record; explain confidentiality, pseudonyms, and right to pause/withdraw.
- Collect brief context: discipline and semester; typical creative coursework; prior exposure to GenAI.

A.2 Screening anchors (2–3 min)

- USAGE FREQUENCY (LAST 3 MONTHS): WEEKLY (3–4 DAYS/WK); DAILY (5–6 DAYS/WK); MULTIPLE-TIMES-DAILY (≥ 2 SESSIONS/DAY ON ≥ 5 DAYS/WK).
- SELF-RATED PROFICIENCY (1–5; 1 = NOVICE, 5 = EXPERT).
- PRIMARY TOOLS (E.G., CHATGPT/GPT-4 CLASS, CLAUDE, BARD; DALL·E, MIDJOURNEY, STABLE DIFFUSION; COPILOT).

A.3 Recent AI episode: ideation (8–10 min)

- Walk through the last time AI was used for ideation.
- Probes: task brief; prompts and iterations; what changed and why; keep/edit/discard decisions; AI role (assistant/partner/critic) and when; effect on originality and confidence; any disclosure (what/how/why)

A.4 Recent AI episode: refinement (6–8 min)

- Describe using AI to refine a concept/draft.
- Probes: requested refinements (structure/clarity/depth/alternatives); evaluation and integration; safeguards against convergence/voice loss; evidence of authorship (rationale notes, drafts, version history).

A.5 Other workflow modes (6–8 min)

- Translation/reformatting (e.g., notes→outline, outline→storyboard): benefits/risks; how intent was preserved.
- Technical assistance (code, data, layout/toolchain): verification steps and handling hallucinations.
- Aesthetic exploration (style prompts, mood boards, variations): preventing cliché/prompt imprinting.
- Concept synthesis: how AI outputs and personal reasoning were combined; criteria for final direction; authorship boundaries.

A.6 Boundaries, authorship, and disclosure (6–8 min)

- Where AI is appropriate vs. not—and why.
- Personal authorship boundary (what remains uniquely the student's).
- Disclosure practices (what, to whom, format, perceived norms, incentives/risks).
- Ownership concerns (dataset/model attribution; reuse rights; comfort with submitting AI-influenced work).

A.7 Impacts on creativity and independence (5–6 min)

- Positives (idea diversity, speed, reduced blank-page anxiety).
- Negatives (convergence, dependency, voice dilution, evaluation difficulty).
- Conditions shaping outcomes (task stage, time pressure, proficiency, assessment criteria).

A.8 Contradictions/negative cases (3–4 min)

- One case where AI unexpectedly reduced originality (why).
- One case where AI markedly improved originality (why).
- What differed (task, prompts, time, stakes).

A.9 Closing (1–2 min)

- One policy change to improve creative learning with AI.
- Anything important not covered; debrief, reminder of confidentiality/withdrawal.